Triphasic Randomized Clinical Trial: Comparison of effects on carbohydrate metabolism

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One hundred thirty women were randomly assigned to treatment with one of three triphasic oral contraceptives, and 43 women using nonhormonal methods served as controls for a 6-month study of the metabolic effects of these formulations. One of the oral contraceptives contained ethinyl estradiol and levonorgestrel, and the other two contained ethinyl estradiol and norethindrone. Compared with pretreatment assessment, all three triphasic oral contraceptives produced small increases in the mean plasma glucose levels that were statistically significant but clinically unimportant. No subjects had abnormal glucose response curves in glucose tolerance test results. Compared with pretreatment assessment, all of the oral contraceptive preparations produced small increases in the mean insulin levels at 3 months but not at 6 months. Overall there were no statistically significant differences among the three formulations in their effects on carbohydrate metabolism as measured by glucose or insulin levels after 6 months of treatment. (AM J OBSTET GYNECOL 1989;161:1402-7.)

Key words: Oral contraceptives, glucose, insulin, comparative trial, triphasic

The first observation that oral contraceptives were associated with abnormal glucose tolerance was published in 1963 by Waine et al.,[1] who noted that men and women with rheumatoid arthritis who took mestranol/norethynodrel (15 to 50 mg/day for 8 to 300 days) had abnormal glucose tolerance test results. Thereafter a number of investigators found that oral contraceptives caused hyperglycemia and hyperinsulinemia. In studies reviewed by Spellacy[2] in 1969 it was presumed that the untoward effect of oral contraceptives on glucose metabolism was caused by the estrogenic component. However, in 1972 Spellacy et al.[3-6] published the first of several studies documenting that progestogens are also responsible for the increased glucose and insulin levels found in women taking oral contraceptives.

A 4-year large-scale survey of glucose tolerance in the United States from 1976 to 1980 found decreased tolerance in 15.4% of oral contraceptives users compared with 6.3% of nonusers.[7] When area under the glucose tolerance curve is used as the index, some impairment of glucose tolerance is found with all combined oral contraceptives. Furthermore, studies that used fixed-dose combination oral contraceptives available at that time found that those containing levonorgestrel were associated with a greater effect on glucose tolerance than those containing norethindrone.[8]

No study has shown a clear-cut relationship between the use of oral contraceptives and precipitation of diabetes. Furthermore a recent long-term study followed more than 100,000 women treated mostly with older, intermediate-dose oral contraceptives for 8 years and found no significant increase in the incidence of cardiovascular disease.[9] Nevertheless the nature and extent of the metabolic effects of oral contraceptives are of considerable interest because impaired glucose tolerance and elevated plasma insulin levels have been linked to an increased risk of coronary heart disease.[10] The triphasic oral contraceptive formulations contain lower doses of steroids than do most of the intermediate-dose fixed-combination oral contraceptives. Because the triphasic preparations are now being widely used, quantitation of their metabolic effects is of considerable interest.

Table I summarizes selected studies of triphasic oral contraceptives on glucose metabolism.[11-18] These studies used serial glucose and insulin measurements after an oral glucose load in women without previous abnormalities of glucose metabolism. These studies show an inconsistent effect of triphasic oral contraceptives on glucose levels in patients who undergo glucose tolerance testing after 3 and 6 months of therapy. In three of the studies[11, 12, 18] there were no significant changes in serum glucose levels. In three studies[13, 15, 16] there were significant increases in only the 30-minute glucose samples. In two studies[14, 17] there were significant increases in all of the glucose values. In none of the studies were there any significant changes in serum insulin values after 3 or 6 months of treatment.

No controlled studies have compared the metabolic effects of triphasic oral contraceptives containing levonorgestrel and norethindrone. In the absence of

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Table I. Effect of triphasic oral contraceptives on carbohydrate metabolism in selected studies*

Report	n	Study duration (mo)	Results of oral glucose tolerance tests	
			Glucose level	Insulin level
Oral contraceptives with levonorgestrel				
Vermuelen and Thiery[11]	5	6	No significant change	No significant change
Skouby et al.[12]†	19	6	No significant change	No significant change
Ellsworth et al.[13]	12	6	Significant increase in 30-min sample only	No significant change
Luyckx et al.[14]	13	6	Significant increase	No significant change
Van der Vange et al.[15]	10	6	Significant increase in 30-min sample only	No significant change
Woutersz et al.[16]	14	6	Significant increase in 30-min sample only	No significant change
Spellacy et al.[17]	29	3	Significant increase	No significant change
Oral contraceptives with norethindrone				
Spellacy et al.[18]	35	3	No significant change	No significant change

*Includes studies that used dynamic testing (oral glucose tolerance) in patients without previous abnormality of glucose metabolism.

†This study also included patients with previous gestational diabetes; however, only normal patients are included in this table.

such studies, theoretic potency considerations and the results of studies conducted with older, higher-dose oral contraceptives have been used to suggest that levonorgestrel-containing triphasic oral contraceptives may have a stronger effect on glucose tolerance than do those containing norethindrone.[5, 8, 19] The Triphasic Randomized Clinical Trial is the first study completed that directly compares the effects of three triphasic oral contraceptives on serum glucose and insulin levels. The numbers of subjects enrolled are higher than for any previous prospective study.

Material and methods

Approximately 50 subjects were randomly assigned to one of three triphasic oral contraceptives; NET (A) (Ortho-Novum 7/7/7), LNG (Tri-Levlen), or NET (B) (Tri-Norinyl). (See Dickerson, these proceedings, for doses and phasing of these formulations.) Forty-nine additional subjects using nonhormonal means of contraception served as the control group. Glucose and insulin data are presented only for those subjects who were evaluated at baseline and at 3 and 6 months.

Subjects were screened for inclusion at the first examination. Inclusion and exclusion criteria were as described (Dickerson, these proceedings). All subjects were required to have a fasting serum glucose level ≤120 mg/dl and were excluded if they had a history of gestational diabetes, had delivered an infant ≥9 pounds, or had a primary relative with diabetes. All subjects gave informed consent.

Oral glucose tolerance tests were conducted between days 17 and 21 of the pretreatment cycle and again between days 17 and 21 of cycles 3 and 6. Subjects were instructed to maintain their usual dietary intake for 3 days before the oral glucose tolerance test, but they were not given a standard caloric loading regimen during this time. The oral glucose tolerance test was preceded by a 14-hour fast. Blood samples were drawn immediately before and 1, 2, and 3 hours after administration of 100 gm of glucose. Activity and external stimuli were not restricted during the oral glucose tolerance test. Glucose and insulin levels for all subjects were determined at a central laboratory. Plasma glucose concentrations were determined with the glucose-HK kit (Boehringer Mannheim Biochemicals, Indianapolis, Indiana) and the Cobas-B10 analyzer (Roche, Basel, Switzerland). Plasma insulin concentrations were determined by radioimmunoassay with the CAC Ria kit (Diagnostic Products Corp., Los Angeles, California).

Because subjects in the control group were not randomized, bias could have been introduced after comparing the differences between the control and the oral contraceptive groups. For example, there were significant differences among the study sites for the control group in areas under the curve of both glucose and insulin. Therefore the control group was excluded from the analysis that involved comparisons among the groups. The control group was included in the tables to show tht normal individuals not taking oral contraceptives are unlikely to have altered carbohydrate metabolism during a 6-month period.

Comparisons were made on the basis of the area under the respective plasma concentration curves of serial glucose and insulin determinations. The insulin index (plasma insulin concentration divided by plasma glucose concentration) was also calculated. In addition, the data were examined to determine if patients had impaired glucose tolerance. Impaired glucose tolerance was defined as a glucose level of ≥200 mg/dl 1 hour after challenge combined with levels ≥140 mg/dl at 2 and 3 hours after challenge.[20]

Table II. Comparative effects of triphasic oral contraceptives on glucose and insulin concentrations after administration of 100 gm of glucose

			Plasma glucose concentration (mg/dl) (mean ± SD)		
Treatment	*Time (hr)*	*n*	*Baseline*	*Cycle 3*	*Cycle 6*
Control	0	43	84.21 ± 11.85	80.35 ± 8.93	82.58 ± 9.38
	1		94.00 ± 23.71	104.00 ± 27.17*	98.30 ± 27.52
	2		85.19 ± 17.41	82.56 ± 22.53	83.23 ± 21.03
	3		74.42 ± 15.57	71.14 ± 14.15	77.19 ± 20.11
NET (A)	0	42	82.29 ± 7.52	80.64 ± 13.30	79.93 ± 6.77
	1		99.29 ± 32.43	120.86 ± 42.79*	111.71 ± 32.23*
	2		83.19 ± 19.27	96.19 ± 33.26*	94.93 ± 25.76*
	3		76.93 ± 16.92	83.74 ± 19.13	82.45 ± 15.82
LNG	0	45	81.91 ± 7.29	81.20 ± 9.06	81.67 ± 6.20
	1		97.58 ± 27.25	110.80 ± 31.88*	116.13 ± 36.19*
	2		87.67 ± 15.92	95.98 ± 26.09*	100.31 ± 22.91*
	3		74.84 ± 18.72	79.22 ± 21.65	81.33 ± 18.31
NET (B)	0	43	81.72 ± 6.93	81.21 ± 11.91	81.07 ± 6.80
	1		99.05 ± 27.76	114.84 ± 33.91*	113.84 ± 33.22*
	2		83.56 ± 18.74	100.51 ± 28.60*	98.67 ± 27.90*
	3		74.56 ± 18.32	81.44 ± 15.44*	85.95 ± 21.78

N, Plasma glucose concentrations for all patients who had evaluations at baseline, cycle 3, and cycle 6. For plasma insulin concentrations there are two additional subjects in the control group, two treated with NET (A), and three treated with NET (B) because of handling errors at one or more examinations.

*$p < 0.05$ (difference of concentrations between visit and baseline for that particular hour).

Table III. Comparative effects of triphasic oral contraceptives on glucose levels

		Area under the curve (mean ± SD) (hr × mg/dl)		
Treatment	*n*	*Baseline*	*Cycle 3*	*Cycle 6*
Control	43	258 ± 43	262 ± 47	261 ± 51
NET (A)	42	262 ± 55	299 ± 77*	288 ± 57*
LNG	45	264 ± 45	287 ± 61*	298 ± 60*
NET (B)	43	261 ± 46	297 ± 62*	296 ± 62*

Note: There were no significant differences among the three oral contraceptives.

*Statistically significant increases were observed ($p < 0.05$) in the area under the curve at both 3- and 6-month visits for all three treatments.

The descriptive statistics of the areas under the curve presented in the tables were computed from the untransformed data, but p values were obtained from the analyses based on the area under the natural logarithm of the curve caused by departures of the assumptions required by analysis of covariance (or analysis of variance for baseline). The analysis of covariance (or analysis of variance for baseline) was performed to compare the overall differences among the treatment groups, with type III sum of squares used for treatment.[21] Then the pairwise comparisons between one treatment group and another were obtained by least squares means in the SAS procedure GLM[21] and were adjusted according to the Bonferroni technique.[22] The paired t test was used to detect the significant changes between baseline and on-treatment values for both the area under the curve and concentration data for the corresponding hours. Significance requires a p value < 0.05.[23]

Results

Table II lists serum glucose and insulin concentrations obtained during glucose tolerance tests. Glucose concentrations are listed for subjects for whom glucose tolerance data were obtained at all three time points (baseline, cycle 3, and cycle 6). The number of subjects in each group who met this requirement was 43 in the control group, 42 for NET (A), 45 for LNG, and 43 for NET (B). For an additional two subjects treated with NET (A), three subjects treated with NET (B), and two subjects in the control group, only insulin data are available because of handling errors at one or more examinations. Glucose data and insulin index data are omitted for these subjects at all points. The inclusion of data from subjects who missed one or the other of the follow-up appointments does not change the results significantly.

Tables III and IV list glucose and insulin values as determined by the area under the curve. Baseline glu-

Treatment	Time (hr)	n	Plasma insulin concentration (μg/dl) (mean ± SD) Baseline	Cycle 3	Cycle 6
Control	0	45	17.27 ± 5.57	16.58 ± 5.73	16.83 ± 5.84
	1		89.24 ± 46.69	85.96 ± 42.00	90.89 ± 67.78
	2		67.51 ± 34.47	63.38 ± 35.71	65.64 ± 35.39
	3		44.42 ± 29.16	36.58 ± 21.87	40.73 ± 24.11
NET (A)	0	44	20.00 ± 12.15	19.84 ± 7.36	17.70 ± 7.87
	1		88.64 ± 42.28	106.73 ± 58.05*	94.89 ± 44.92
	2		67.48 ± 35.16	92.27 ± 72.10*	78.93 ± 42.77*
	3		46.93 ± 24.76	66.73 ± 54.88*	50.95 ± 30.57
LNG	0	45	20.62 ± 9.11	23.11 ± 11.50	18.49 ± 9.04
	1		99.09 ± 75.70	100.53 ± 49.21	97.18 ± 61.11
	2		80.13 ± 83.50	84.64 ± 51.53	85.69 ± 58.86
	3		47.44 ± 47.78	61.13 ± 81.36*	55.51 ± 39.05
NET (B)	0	46	20.91 ± 9.13	23.11 ± 10.41	18.15 ± 6.98
	1		93.09 ± 39.69	101.09 ± 47.87	92.98 ± 45.56
	2		78.87 ± 45.63	91.17 ± 47.41	78.43 ± 42.34
	3		47.37 ± 29.27	67.59 ± 44.56*	59.43 ± 48.39

Table IV. Comparative effects of triphasic oral contraceptives on insulin levels

Treatment	n	Area under the curve (mean ± SD) (hr × μg/dl) Baseline	Cycle 3	Cycle 6
Control	45	188 ± 87	176 ± 73	185 ± 100
NET (A)	44	190 ± 79	242 ± 142*	208 ± 93
LNG	45	213 ± 173	227 ± 130	220 ± 130
NET (B)	46	206 ± 86	238 ± 102*	210 ± 89

Note: There were no statistically significant differences among the three oral contraceptives.

*Statistically significantly different from baseline, $p < 0.05$.

Table V. Plasma glucose levels: mean change from baseline area under the curve

Treatment	n	Cycle 3 (hr × mg/dl)	Cycle 6 (hr × mg/dl)
Control	43	3.8 ± 51.8	2.9 ± 37.9
NET (A)	42	37.2 ± 70.2	25.8 ± 59.2
LNG	45	23.4 ± 52.6	34.3 ± 65.6
NET (B)	43	35.9 ± 63.7	35.3 ± 60.8

Table VI. Plasma insulin levels: mean change from baseline area under the curve

Treatment	n	Cycle 3 (hr × μg/dl)	Cycle 6 (hr × μg/dl)
Control	45	−11.7 ± 98.9	−2.3 ± 91.9
NET (A)	44	52.7 ± 121.9	18.6 ± 71.5
LNG	45	14 ± 98.8	6.6 ± 111.0
NET (B)	46	31.5 ± 107.4	4.1 ± 87.2

cose and insulin values, whether expressed as serum concentrations or as areas under the curve, were similar for all treatment groups (Tables II, III, and IV). All three oral contraceptive formulations produced similar, small, significant increases in postchallenge glucose levels at the 3- and 6-month examinations compared with baseline.

Greater variability was seen in the effect on insulin levels (Tables II and IV). Small but significant increases in serum insulin levels at the 3-month, but not the 6-month, examination were observed in all treatment groups (Table III). No significant differences were seen among oral contraceptive regimens in their effect on glucose and insulin values as determined by changes from baseline at either 3 or 6 months (Tables V and VI). Because of differences in baseline areas under the curve among the groups, the comparison is based on the change in the area under the curve with each drug rather than on the area under the curve itself. The data were also calculated on the basis of the insulin index (concentration of insulin divided by concentration of glucose as a rough measure of "insulin resistance"), with similar results (Table VII). Notably, no subject in the study had an abnormal glucose tolerance curve during any examination.

Comment

The data in this study corroborate earlier reports that low-dose triphasic oral contraceptives have only minor effects on glucose tolerance.[11-18] The increases in glucose values were small and did not increase significantly with time after the initial 3 months of use. Even less effect was seen on insulin levels, which are believed to be associated more closely with the risk of coronary heart disease.[10] LNG showed no statistically significant change in insulin levels at either examination. Although

Table VII. Comparative effects of triphasic oral contraceptives on insulin index*

Treatment	*n*	*0 Hour*	*1 Hour*	*2 Hours*	*3 Hours*
Control					
Baseline	43	0.21	0.93	0.79	0.58
Cycle 3		0.21	0.81	0.74	0.50
Cycle 6		0.20	0.88	0.77	0.53
NET (A)					
Baseline	42	0.23	0.93	0.82	0.60
Cycle 3		0.26	0.92	0.95	0.75
Cycle 6		0.22	0.88	0.82	0.60
LNG					
Baseline	45	0.25	1.01	0.88	0.61
Cycle 3		0.29	0.94	0.89	0.73
Cycle 6		0.23	0.84	0.83	0.67
NET (B)					
Baseline	43	0.25	0.95	0.91	0.60
Cycle 3		0.28	0.92	0.93	0.80
Cycle 6		0.22	0.84	0.76	0.65

*Insulin index = insulin concentration/glucose concentration.

both NET (A) and NET (B) had statistically significant increases from baseline at 3 months, this was not observed after 6 months of treatment. The clinical significance of these observations is unknown. The most important result is that no subjects in any treatment group had significantly impaired glucose tolerance, as defined by an abnormal glucose curve. The results of this study do not apply to patients with gestational diabetes. In a study with an insulin clamp technique in subjects with a history of gestational diabetes, Skouby et al.[24] have demonstrated decreased insulin sensitivity after 6 months of treatment with a levonorgestrel-containing triphasic oral contraceptive.

The data strongly suggest that the predictions that levonorgestrel-containing triphasic oral contraceptives would have a greater adverse effect on carbohydrate metabolism in normal subjects than those oral contraceptives containing norethindrone were incorrect.[18]

In conclusion, low-dose triphasic oral contraceptives produce minor increases in plasma glucose and insulin levels in normal women, as detected by oral glucose tolerance test results that are statistically significant but clinically unimportant. The results of this randomized clinical trial suggest that in normal women the choice of triphasic oral contraceptives should be based on considerations other than their effects on carbohydrate metabolism.

REFERENCES

1. Waine H, Frieden EH, Caplan HI, Cole T. Metabolic effects of Enovid in rheumatoid patients. Arthritis Rheum 1963;6:796.
2. Spellacy WN. A review of carbohydrate metabolism and the oral contraceptives. Am J Obstet Gynecol 1969; 140:448-60.
3. Spellacy WN, Buhi WC, Birk SA. The effect of the progestogen ethynodiol, diacetate on glucose, insulin, and growth hormone after six months' treatment. Acta Endocrinol 1972;70:373-84.
4. Spellacy WN, Buhi WC, Birk SA, McCreary SA. Metabolic studies in women on norethindrone for 6 months' time (measurements of blood glucose, insulin, and triglyceride concentrations). Fertil Steril 1973;24:419-25.
5. Spellacy WN, Buhi WC, Birk SA. Norgestrel and carbohydrate-lipid metabolism: glucose, insulin and triglyceride changes during six months' time use. Contraception 1974;9:615-25.
6. Spellacy WN, Buhi WG, Birk SA. Effects of norethindrone on carbohydrate and lipid metabolism. Obstet Gynecol 1975;46:560-3.
7. Russell-Briefel R, Ezzati TM, Perlman JA, Murphy RS. Impaired glucose tolerance in women using oral contraceptives: United States, 1976-1980. J Chron Dis 1987; 40:3-11.
8. Wynn V, Adams PW, Godland IF, et al. Comparison of effects of different combined oral-contraceptive formulations on carbohydrate and lipid metabolism. Lancet 1979;1:1045-9.
9. Stampfer MJ, Willett WC, Colditz GA, Speizer FE, Hennekens CH. A prospective study of past use of oral contraceptive agents and risk of cardiovascular diseases. N Engl J Med 1988;319:1313-7.
10. Crook D. Godsland IF, Wynn V. Oral contraceptive and coronary heart disease: modulation of glucose tolerance and plasma lipid risk factors by progestins. Am J Obstet Gynecol 1988;158:1612-20.
11. Vermeulen A, Thiery M. Metabolic effects of the triphasic oral contraceptive Trigynon. Contraception 1982;26:505-13.
12. Skouby SO, Kuehl C, Molsted-Pedersen L, Petersen K, Christensen MS. Triphasic oral contraception: metabolic effects in normal women and those with previous gestational diabetes. Am J Obstet Gynecol 1985;153:495-500.
13. Ellsworth HS, Ayerst RI, Harris JW, et al. Evaluation of new triphasic oral contraceptive in private practice. Contraception 1986;34:435-43.
14. Luyckx AS, Gaspard UJ, Romus MA, Grigorescu F, De Meyts P, Lefèbvre P. Carbohydrate metabolism in women who used oral contraceptives containing levonorgestrel or desogestrel: a 6-month prospective study. Fertil Steril 1986;45:635-42.
15. Van der Vange, Kloosterboer HJ, Haspels AA. Effect of seven low-dose combined oral contraceptive preparations on carbohydrate metabolism. Am J Obstet Gynecol 1987;156:918-22.
16. Woutersz TB, Butler AJ, Cohen M, et al. A low-dose triphasic oral contraceptive. Fertil Steril 1987;47:425-30.
17. Spellacy WN, Ellingson AB, Kotlik A, Tsibris JC. Plasma glucose and insulin levels in women using a

levonorgestrel-containing triphasic oral contraceptive for three months. Contraception 1988;38:27-34.
18. Spellacy WN, Ellingson AB, Kotlik A, Tsibris JCM. Prospective study of carbohydrate metabolism in women using a triphasic oral contraceptive containing norethindrone and ethinyl estradiol for three months. AM J OBSTET GYNECOL 1988;159:877-9.
19. Wynn V, Godsland I. Effects of oral contraceptives on carbohydrate metabolism. J Reprod Med 1986;31:892-7.
20. National Diabetes Data Group. Classification and diagnosis of diabetes mellitus and other categories of glucose intolerance. Diabetes 1979;28:1039-57.
21. SAS Institute, Inc. SAS user's guide: statistics. 5th ed. Cary, NC: SAS Institute, Inc, 1985.
22. Miller RG Jr. Simultaneous statistical inference. New York: Springer-Verlag, 1981.
23. Snedecor GW, Cochran WG. Statistical methods, 7th ed. Cedar Rapids, IA: Iowa State University Press, 1980.
24. Skouby SO, Andersen O, Sourbrey N, Kuhl C. Oral contraception and insulin sensitivity: in vivo assessment in normal women and women with previous gestational diabetes. J Clin Endocrinol Metab 1987;64:519-23.

Triphasic Randomized Clinical Trial: Comparative frequency of intermenstrual bleeding

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Oral contraceptives have the highest theoretic efficacy of all reversible forms of contraception, yet in practice they have an actual failure rate equivalent to that of some barrier methods. The efficacy of oral contraceptives depends primarily on patient compliance. So-called "nuisance" side effects, such as intermenstrual bleeding (breakthrough bleeding or spotting), play a major role in their discontinuation. The Triphasic Randomized Clinical Trial compared the incidence of these side effects in three groups of subjects taking NET (A) (Ortho-Novum 7/7/7), LNG (Tri-Levlen), or NET (B) (Tri-Norinyl). Subjects taking NET (A) had the highest incidence of intermenstrual bleeding (63%) followed by NET (B) (44%) and LNG (33%). Only the difference between NET (A) and LNG attained statistical significance ($p < 0.030$). Control subjects had no breakthrough bleeding and a 4% incidence of spotting. Further research is warranted to establish the cause of the disparity in the incidence of intermenstrual bleeding among patient groups assigned to the three oral triphasic contraceptives. (AM J OBSTET GYNECOL 1989;161: 1407-11.)

Key words: Oral contraceptives, intermenstrual bleeding, breakthrough bleeding, adverse effects, comparative trials

Oral contraceptives have the highest theoretic effectiveness of all reversible methods of contraception. In practice, however, this method has an actual failure rate equivalent to that of some barrier methods. Given that the efficacy of oral contraceptives, especially that of the lower dose products, depends on good patient compliance,[1] this discrepancy may result from the high incidence of nuisance side effects. Such side effects can cause poor compliance or discontinuation of the regimen altogether.[2] These adverse effects include intermenstrual bleeding (breakthrough bleeding or spotting), amenorrhea (failure of withdrawal bleeding), nausea, vomiting, edema, mood changes, and weight gain.

Few randomized clinical trials have been performed to date to compare the incidence of such side effects among triphasic oral contraceptives. This multicenter, outpatient, randomized, open-label 6-month study with three currently available triphasic oral contraceptive formulations provides data concerning bleeding abnormalities.

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Material and methods

Study groups. Two hundred six female subjects participated in the triphasic randomized clinical trial, with 157 women randomly assigned to receive triphasic reg-